

Patient Health Monitoring Pipeline Using AWS

PROJECT TITLE

Patient Health Monitoring Data Pipeline

BUSINESS SCENARIO

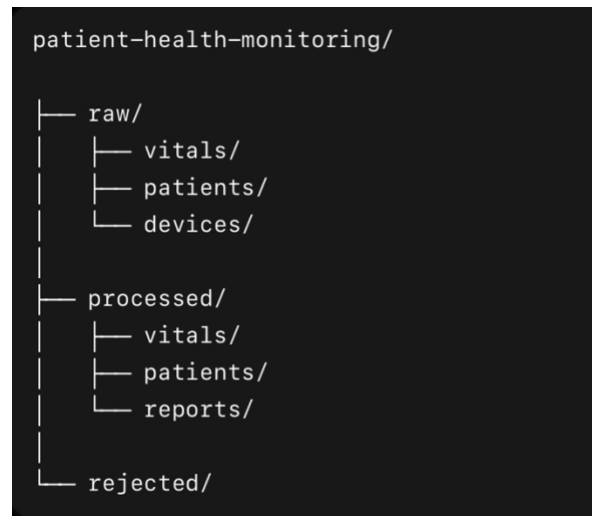
A hospital receives patient health monitoring files every hour from:

- ICU monitoring systems
- Wearable health devices
- Hospital management systems
- Emergency care systems

The hospital wants to:

- Store incoming patient data securely
- Validate health records automatically
- Detect corrupted files
- Clean and transform patient data
- Catalog metadata automatically
- Store processed healthcare data
- Separate bad patient records
- Maintain production-ready ETL workflows

PROJECT ARCHITECTURE



PHASE 1 — PROJECT DEFINITION

Objective

Build a healthcare ETL pipeline that processes patient monitoring data securely.

PROJECT GOALS

build a system that:

- Ingests patient data
- Validates incoming files
- Cleans medical records
- Detects invalid patient readings
- Stores transformed healthcare data
- Handles schema changes

HEALTH METRICS TO TRACK

Metric	Example
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Heart Rate	72 BPM
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Blood Pressure	120/80
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Oxygen Level	98%
Temperature	98.6 F
Respiration Rate	16/min

PHASE 2 — DATA COLLECTION & STORAGE

TASK 1 — Create S3 Bucket

Bucket Name

patient-health-monitoring

Create Folder Structure

raw/vitals/

raw/patients/

raw/devices/

processed/vitals/

processed/patients/

processed/reports/

rejected/

PHASE 3 — IAM IMPLEMENTATION

TASK 1 — Glue IAM Role

Permissions

- * Read raw bucket
- * Write processed bucket
- * Access Glue Catalog
- * Access logs

TASK 2 — Lambda IAM Role

Permissions

- * Read uploaded files
- * Move rejected files
- * Trigger crawler
- * Trigger ETL job

PHASE 4 — SAMPLE HEALTHCARE DATA

FILE 1 — patient_vitals.csv

patient_id,timestamp,heart_rate,blood_pressure,oxygen_level,temperature

P1001,2025-01-10 08:00:00,72,120/80,98,98.6

P1002,2025-01-10 08:05:00,95,140/90,96,99.1

P1003,2025-01-10 08:10:00,45,90/60,88,101.2

P1004,2025-01-10 08:15:00,,130/85,97,98.4

P1005,2025-01-10 08:20:00,110,150/95,,100.1

P1005,2025-01-10 08:20:00,110,150/95,,100.1

FILE 2 — patient_details.csv

patient_id,name,age,gender,city,disease

P1001,Rahul Sharma,45,Male,Delhi,Diabetes

P1002,Priya Verma,52,Female,Mumbai,Hypertension

P1003,Amit Singh,60,Male,Pune,Heart Disease

P1004,Neha Kapoor,29,Female,Delhi,Asthma

P1005,Rohit Jain,48,Male,Jaipur,Diabetes

FILE 3 — device_logs.csv

device_id,patient_id,device_type,status,last_sync

D101,P1001,SmartWatch,Active,2025-01-10 08:00:00

D102,P1002,Oximeter,Inactive,2025-01-10 07:50:00

D103,P1003,HeartMonitor,Active,2025-01-10 08:05:00

REAL-WORLD DATA PROBLEMS INCLUDED

Students must detect:

Issue	Example	
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Null values	heart_rate missing	
Duplicate records	P1005 repeated	
Critical health readings	oxygen_level = 88	
Missing patient metrics	oxygen_level blank	
Schema changes	new column added	

PHASE 5 — GLUE CRAWLER

Objective

Automatically discover healthcare schema.

TASKS

Students must:

- * Create crawler
- * Scan raw healthcare bucket
- * Create Glue database: healthcare_db

Expected Tables

Table	Source
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patient_vitals	raw/vitals
patient_details	raw/patients
device_logs	raw/devices

PHASE 6 — GLUE DATA CATALOG

Objective

Store metadata automatically.

Catalog contains:

- * Column names
- * Data types
- * Table locations

PHASE 7 — GLUE ETL JOB

MAIN HEALTHCARE ETL PIPELINE

Students must create Glue ETL using PySpark.

ETL REQUIREMENTS

Step 1 — Read Raw Files

Read:

- * patient_vitals.csv
- * patient_details.csv
- * device_logs.csv

Step 2 — Data Cleaning

Handle:

- * Null readings
- * Duplicate records
- * Incorrect values
- * Missing patient information

Step 3 — Data Transformation

Create:

- * health_status
- * alert_flag
- * monitoring_date

* monitoring_hour

Example Rules

Condition	Status
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oxygen_level < 90	Critical
temperature > 100	Fever
heart_rate > 100	High Risk

Step 4 — Join Datasets

Join:

* Patient vitals

* Patient details

* Device logs

Final Output Columns

Column

patient_id
patient_name
disease
heart_rate
oxygen_level

| temperature |
| health_status |
| device_type |
| monitoring_date |

Step 5 — Store Processed Data

Write transformed data into:

processed/reports/

Format:

* Parquet

PHASE 8 — EXPECTED OUTPUT SAMPLE

patient_id,patient_name,disease,heart_rate,oxygen_level,temperature,health_status

P1001,Rahul Sharma,Diabetes,72,98,98.6,Normal

P1002,Priya Verma,Hypertension,95,96,99.1,Warning

P1003,Amit Singh,Heart Disease,45,88,101.2,Critical